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A Note on

A toy model of two countries

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1. Economic question and literature

A productivity shock in one country changes its output, its demand for foreign goods and the return to saving. Its international effects depend on relative goods prices, the currency value of outstanding assets and the degree to which households can share risk. The economic question is therefore not simply whether two outputs move together, but which trade and financial mechanisms can explain that comovement and the accompanying real exchange-rate movement.

Backus et al. [1992] establish the international real-business-cycle benchmark and its implications for cross-country quantities. The two-country structure here retains distinct home and foreign production and imperfectly substitutable consumption goods. Country sizes matter because a unit of per-capita exports from one economy becomes imports spread over the population of its partner. Confusing per-capita quantities with world totals would change the resource constraints even when prices are unchanged.

With a single international bond, net foreign assets carry history into future consumption. Schmitt-Grohé and Uribe [2003] compare stationarity-inducing closures for incomplete-market open economies. A small portfolio adjustment wedge is used here to anchor net foreign assets locally. Its role is different from perfect international risk sharing. Obstfeld and Rogoff [1995] connects the intertemporal open-economy framework with nominal rigidities and exchange-rate dynamics. The original note's nominal extension is retained below, separately from the real simulation starter. Quadratic price adjustment costs follow Rotemberg [1982], and interest-rate policy follows the rule-based approach of Taylor [1993].

The assessment asks students to connect domestic and foreign quantities, relative prices and asset positions with empirical measurements. A successful simulation is not evidence that one country's output series can stand in for an entire currency area, nor that a nominal bilateral exchange rate is already the model's real exchange rate.

2. The two-country real economy

The countries are H and F, of population shares n and $1-n$. Upper-case quantities below are per-capita levels. Let e_t be home currency per unit of foreign currency, $P_{i,t}$ the producer price and $P^C_{i,t}$ the consumption price index. Define

$$rer_t = e_t P^C_{F,t}/P^C_{H,t}, \qquad p_{H,t} = P_{H,t}/P^C_{H,t}, \qquad p_{F,t} = P_{F,t}/P^C_{F,t}. \tag{1}$$

An increase in rer_t is a home real depreciation. The import price in home consumption units is $rer_t p_{F,t}$, not rer_t alone. The foreign import price is $p_{H,t}/rer_t$.

2.1 Households

Households in country i maximize

$$\max_{\{C_{i,t+\tau},D_{i,t+\tau},B_{i,t+\tau},H_{i,t+\tau}\}_{\tau\geq 0}} E_t \sum_{\tau\geq 0} \beta_i^\tau \left[\frac{(C_{i,t+\tau} - X_{i,t+\tau})^{1-\sigma_C}}{1-\sigma_C} - \chi_i \frac{H_{i,t+\tau}^{1+\sigma_H}}{1+\sigma_H} \right]. \tag{2}$$

The original external-habit possibility is $X_{i,t} = hC_{i,t-1}$, taken as given individually. The real starter sets $h = 0$. This is a specialization of the preferences, not an omitted habit equation. The budget can be written in nominal home units to avoid mixing asset denominations:

$$P^C_{H,t} C_{H,t} + D_t + e_t B_t + \mathcal{A}_t = W_t H_{H,t} + \Pi_t + R^N_{H,t-1} D_{t-1} + e_t R^N_{F,t-1} B_{t-1}. \tag{3}$$

D_t is a domestic nominal bond and B_t a foreign-currency bond. The cost $\mathcal{A}_t$ is a portfolio adjustment expenditure. It is zero and has zero first-order resource effect at its steady-state target. Its marginal wedge is retained in asset pricing.

Let $\mathcal{B}_{i,t}$ be the real budget residual, income minus expenditure. With utility $U_{i,t}$, the Lagrangian is

$$\mathcal{L}_{i,t} = E_t \sum_{\tau\geq 0} \beta_i^\tau [U_{i,t+\tau} + \lambda^C_{i,t+\tau} \mathcal{B}_{i,t+\tau}].$$

Its consumption and labor conditions, and the domestic nominal bond condition, are

$$\lambda^C_{i,t} = (C_{i,t} - X_{i,t})^{-\sigma_C}, \tag{4}$$

$$\chi_i H_{i,t}^{\sigma_H} = \lambda^C_{i,t} w_{i,t}, \tag{5}$$

$$\lambda^C_{i,t} = \beta_i R^N_{i,t} E_t(\lambda^C_{i,t+1}/\pi^C_{i,t+1}). \tag{6}$$

Here $w_{i,t} = W_{i,t}/P^C_{i,t}$ and $\pi^C_{i,t+1} = P^C_{i,t+1}/P^C_{i,t}$. An expectation of a ratio cannot in general be replaced by a ratio of expectations. For a foreign bond, writing Ψ_t for its marginal portfolio wedge, the home condition is

$$\lambda^C_{H,t}(1+\Psi_t) = \beta_H R^N_{F,t} E_t \left[\lambda^C_{H,t+1} \frac{e_{t+1}/e_t}{\pi^C_{H,t+1}} \right]. \tag{7}$$

The real starter uses the first-order counterparts with gross real returns $r_{H,t}$ and $r_{F,t}$. Around the target, the wedge is $\Psi_t = \chi_B(b_{H,t} - \bar{b}_H) - e^E_t$ to first order. Here e^E_t denotes the portfolio disturbance, not the nominal exchange rate e_t.

2.2 Consumption goods and trade

The constant-elasticity-of-substitution (CES) composite at home is

$$C_{H,t} = \left[(1-\alpha_C)^{1/\mu} c_{HH,t}^{(\mu-1)/\mu} + \alpha_C^{1/\mu} c_{HF,t}^{(\mu-1)/\mu}\right]^{\mu/(\mu-1)}. \tag{8}$$

The analogous foreign composite reverses the goods. The parameter $\mu > 0$ is the substitution elasticity, with the Cobb–Douglas limit at $\mu = 1$. The dual price identities are

$$1 = (1-\alpha_C) p_{H,t}^{1-\mu} + \alpha_C (rer_t p_{F,t})^{1-\mu}, \tag{9}$$

$$1 = (1-\alpha_C) p_{F,t}^{1-\mu} + \alpha_C (p_{H,t}/rer_t)^{1-\mu}. \tag{10}$$

Without demand wedges, optimal quantities are

$$c_{HH,t} = (1-\alpha_C) p_{H,t}^{-\mu} C_{H,t}, \qquad c_{HF,t} = \alpha_C (rer_t p_{F,t})^{-\mu} C_{H,t}, \tag{11}$$

$$c_{FF,t} = (1-\alpha_C) p_{F,t}^{-\mu} C_{F,t}, \qquad c_{FH,t} = \alpha_C (p_{H,t}/rer_t)^{-\mu} C_{F,t}. \tag{12}$$

The starter multiplies imports $c_{HF,t}$ by $\varepsilon_{H,t}^X$ and $c_{FH,t}$ by $\varepsilon_{F,t}^X$. These are reduced-form import-demand wedges indexed by the importing country. They are not time-varying CES weights inside the displayed composite, which would also alter the exact price index. Consequently the wedge version is a local demand system, not a fully specified nonlinear preference model. This limitation must be retained in empirical interpretation. Home exports depend on the foreign import wedge.

2.3 Production

The real benchmark has competitive producers with decreasing returns to labor:

$$Y_{i,t} = \varepsilon_{i,t}^Z H_{i,t}^{\alpha}. \tag{13}$$

There is no capital input in this production function. Profits in local currency are $\Pi_{i,t} = P_{i,t} Y_{i,t} - W_{i,t} H_{i,t}$. Cost minimization gives

$$MC_{i,t} = \frac{W_{i,t} H_{i,t}}{\alpha Y_{i,t}}. \tag{14}$$

Competition sets $MC_{i,t} = P_{i,t}$, so in consumption units $w_{i,t} = p_{i,t} \alpha Y_{i,t}/H_{i,t}$. Profits are rebated to households. The relative producer price must enter the real wage equation.

2.4 Resource constraints and asset accumulation

Ignoring second-order portfolio costs in the linear benchmark,

$$Y_{H,t} = (1-\alpha_C) p_{H,t}^{-\mu} C_{H,t} + \frac{1-n}{n} \varepsilon_{F,t}^X \alpha_C (p_{H,t}/rer_t)^{-\mu} C_{F,t}, \tag{15}$$

$$Y_{F,t} = (1-\alpha_C) p_{F,t}^{-\mu} C_{F,t} + \frac{n}{1-n} \varepsilon_{H,t}^X \alpha_C (rer_t p_{F,t})^{-\mu} C_{H,t}. \tag{16}$$

These are per-capita resource constraints. World home exports are $(1-n)c_{FH,t}$ and world foreign exports are $nc_{HF,t}$. Domestic bonds are in zero net supply. Define $b_{H,t} = e_t B_t / P_{H,t}^C$, the value of home net foreign assets in home consumption units. Aggregating the nominal household budget and profits gives

$$e_t B_t - e_t R_{F,t-1}^N B_{t-1} = P_{H,t} Y_{H,t} - P_{H,t}^C C_{H,t} - \mathcal{A}_t.$$

After dividing by the consumption price index,

$$b_{H,t} = r_{F,t-1}\frac{rer_t}{rer_{t-1}}b_{H,t-1} + p_{H,t}Y_{H,t} - C_{H,t} - a_t, \tag{17}$$

where $a_t = \mathcal{A}_t/P^C_{H,t}$ and the foreign real return is measured using the foreign consumption deflator. The factor rer_t/rer_{t-1} revalues inherited foreign-currency assets. It is a wealth gain for a creditor and a loss for a debtor when home depreciates.

If foreign assets are measured in foreign consumption units, international clearing requires

$$nb_{H,t} + (1-n)rer_t b_{F,t} = 0. \tag{18}$$

The exchange-rate conversion is essential away from $\overline{rer} = 1$. Negative net foreign assets are represented in code as proportional deviations $(b_t - \bar{b})/\bar{b}$, not as the logarithm of a negative stock.

3. Stationarity and the real simulation starter

Relative prices remove nominal unit roots, but stationarity of net foreign assets additionally requires a financial closure. The first-order domestic Euler equations and foreign-bond condition imply

$$E_t\widehat{rer}_{t+1} - \widehat{rer}_t = \hat{r}_{H,t} - \hat{r}_{F,t} - e^E_t + \chi_B\bar{b}_H\hat{b}_{H,t}. \tag{19}$$

This is a local uncovered-interest-parity relation, not an exact nonlinear expectation identity. The quadratic portfolio cost is a convenient way to obtain a marginal wedge linear in the asset deviation. Asset-pricing risk covariances are absent at first order.

In `soe_RBC.mod`, lower-case variables are log deviations, except the proportional asset deviations just defined. The two CPI equations are

$$0 = (1-\alpha_C)\hat{p}_H + \alpha_C(\hat{p}_F + \widehat{rer}), \qquad 0 = (1-\alpha_C)\hat{p}_F + \alpha_C(\hat{p}_H - \widehat{rer}).$$

The model uses $\sigma_C = 1.5$, $\sigma_H = 2$, $\alpha = .7$, $\chi_B = .007$, $\mu = 2$, $\alpha_C = .1$, $n = .4$ and $\bar{r}_F = 1/.994$. Both steady-state labor supplies are 1/3. Consumption and asset positions satisfy the two countries' resource constraints and budget balances, rather than being imposed equal across countries. Productivity persistence is .95, import-wedge persistence .4, and portfolio-shock persistence .1.

Five innovation processes are declared. The starter deliberately assigns standard deviation .01 only to home productivity, home import demand and the portfolio disturbance. Foreign innovations therefore have zero variance in this baseline. This matters for stochastic singularity if many observables are added. No shock variances or measurement errors should be silently added to make an estimation run.

4. Nominal extension: prices and policy

The original note also describes a nominal extension. The following retailer, price-setting and authority blocks are retained as theory, not as equations already implemented by the real starter. Completing this extension requires adding its nominal variables, both policy rules and the corresponding market clearing terms. It does not require deleting the real benchmark.

4.1 Retailers

Competitive retailers combine intermediate varieties $v \in [0,1]$:

$$Y_t = \left[\int_0^1 y_{v,t}^{(\epsilon-1)/\epsilon}\, dv\right]^{\epsilon/(\epsilon-1)}, \qquad \epsilon > 1. \tag{20}$$

They maximize $P_t Y_t - \int_0^1 P_{v,t} y_{v,t}\, dv$ subject to this technology. Cost minimization and zero profits imply

$$y_{v,t} = (P_{v,t}/P_t)^{-\epsilon} Y_t, \tag{21}$$

$$P_t = \left[\int_0^1 P_{v,t}^{1-\epsilon}\, dv\right]^{1/(1-\epsilon)}. \tag{22}$$

The competitive limit is $\epsilon \to \infty$, not $\epsilon \to 1$.

4.2 Intermediate firms and price adjustment

Each intermediate producer uses $y_{v,t} = \varepsilon_t^Z H_{v,t}^{\alpha}$. Its marginal cost is $MC_{v,t} = W_t H_{v,t}/(\alpha y_{v,t})$. A price change uses a fraction

$$\Delta_{v,t}^P = \frac{\chi_P}{2}(P_{v,t}/P_{v,t-1} - 1)^2 \tag{23}$$

of aggregate output. With $mc_t = MC_t/P_t$, the price-setting objective is

$$\max_{\{P_{v,t+\tau}\}_{\tau\geq 0}} E_t \sum_{\tau\geq 0} \beta^\tau \frac{\lambda_{t+\tau}^C}{\lambda_t^C} p_{t+\tau} Y_{t+\tau} \left[\left(\frac{P_{v,t+\tau}}{P_{t+\tau}}\right)^{1-\epsilon} - \varepsilon_{t+\tau}^P mc_{t+\tau} \left(\frac{P_{v,t+\tau}}{P_{t+\tau}}\right)^{-\epsilon} - \Delta_{v,t+\tau}^P\right]. \tag{24}$$

Under symmetry, producer inflation $\pi_t = P_t/P_{t-1}$ satisfies

$$\begin{aligned} 0 = {} & 1 - \epsilon + \epsilon \varepsilon_t^P mc_t - \chi_P \pi_t(\pi_t - 1) \\ & + \beta E_t\left[\frac{\lambda_{t+1}^C}{\lambda_t^C} \frac{p_{t+1} Y_{t+1}}{p_t Y_t} \chi_P \pi_{t+1}(\pi_{t+1} - 1)\right]. \end{aligned} \tag{25}$$

The factor p_{t+1}/p_t is needed because the discount factor is in consumption units while adjustment costs are paid in producer goods. The disturbance ε_t^P is a markup wedge, distinct from productivity.

4.3 Authorities and nominal stationarity identities

In each country the nominal policy rate follows

$$R_t^N = (R_{t-1}^N)^{\rho_R}\left[\bar{R}^N(\pi_t^C/\bar{\pi}^C)^{\phi_\pi}(Y_t/\bar{Y})^{\phi_y}\right]^{1-\rho_R} \varepsilon_t^R. \tag{26}$$

The parameters measure smoothing and responses to inflation and output. The output ratio is a deviation from steady state, not automatically a welfare-relevant output gap. The stationarity identities are

$$p_t/p_{t-1} = \pi_t/\pi_t^C, \tag{27}$$

$$rer_t/rer_{t-1} = (e_t/e_{t-1})\pi_{F,t}^C/\pi_{H,t}^C. \tag{28}$$

The foreign exchange rates are the reciprocals of the home rates. The exact foreign-bond Euler condition was given above. Its first-order nominal counterpart is

$$E_t\widehat{\Delta e}_{t+1} = \widehat{R}_{H,t}^N - \widehat{R}_{F,t}^N + \Psi_t.$$

Together with CPI inflation this delivers the real relation in the starter. It must not be presented as an exact nonlinear division of Euler expectations.

4.4 Nominal resource and financial accounting

The home resource constraint now includes price and portfolio costs:

$$Y_{H,t} = (1-\alpha_C)p_{H,t}^{-\mu}C_{H,t} + \frac{1-n}{n}\varepsilon_{F,t}^{X}\alpha_C(p_{H,t}/rer_t)^{-\mu}C_{F,t} + \Delta_{H,t}^{P}Y_{H,t} + a_{H,t}/p_{H,t}. \tag{29}$$

The foreign equation is obtained by swapping countries and replacing n with $1-n$. In integral form, the home demand aggregates households over $[0,n]$ and foreign importers over $[n,1]$. The foreign demand aggregates foreign households over $[n,1]$ and home importers over $[0,n]$, never over $[1,n]$. Net foreign assets obey

$$b_{H,t} = r_{F,t-1}\frac{rer_t}{rer_{t-1}}b_{H,t-1} + p_{H,t}(1-\Delta_{H,t}^{P})Y_{H,t} - C_{H,t} - a_{H,t}. \tag{30}$$

Substituting the resource constraint cancels the two domestic adjustment-cost terms, leaving the value of goods sold to domestic and foreign consumers minus home consumption. This reproduces the original trade-balance derivation while keeping currencies, price deflators and the timing of inherited assets explicit.

5. Empirical starting point

The downloader supplies raw US and French output, not US and euro-area output. No exchange-rate series is included. Document the country pair, nominal exchange-rate convention, consumption deflators, common dates and trade-flow units. The calibration $n = .4$ is a model parameter, not a documented estimate of the relative size of that empirical pair. Real exchange rates, outputs and exports require distinct measurement choices. Data transformations, priors, estimated shock scales and posterior results remain assessment work.

References

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